

Bijan Pourriahi

Senior Full-Stack / Platform Engineer

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PROFILE

Senior full-stack / platform engineer with 10+ years of experience building production software across SaaS, fintech, analytics, automation, market research, and data-heavy systems. Strong across frontend applications, backend services, APIs, databases, data modeling, infrastructure, DevOps, monitoring, deployment workflows, and internal tooling.

Built production systems for startups, SaaS companies, digital agencies, and a hedge fund, including Rails/PostgreSQL applications, Python data pipelines, operational dashboards, trading/research infrastructure, automation tools, and cloud deployment systems. Strongest in roles requiring end-to-end ownership across product, backend, data, and infrastructure.

CORE SKILLS

Full-stack engineering: React, TypeScript, Rails, FastAPI, APIs, dashboards, admin tools, browser extensions, WebSockets, background jobs.

Data systems: PostgreSQL, ClickHouse, SQLite, data modeling, ETL, event logs, replayable workflows, analytics pipelines, query optimization.

Platform / infrastructure: Kubernetes, Docker, AWS, Linux, Ansible, CI/CD, Grafana, monitoring, deployment automation, sysadmin, production operations.

Systems ownership: architecture, debugging, performance tuning, operational tooling, automation, product/backend/data/infra tradeoffs.

PROFESSIONAL EXPERIENCE

QuantBox — Principal Full-Stack / Platform Engineer

2021-Present | Remote

- Built a Rust/Python research and operations platform covering ingestion, replay, simulation, evaluation, runtime systems, evidence capture, and execution tooling.
- Built an operator-facing workstation for managing multiple real-time accounts from one interface, including account state, fan-out controls, risk limits, kill switches, flatten/cancel controls, and monitoring views.
- Designed replayable workflows for testing system behavior across live, historical, and simulated data sources.
- Built analytics and intelligence tooling using SQLite, ClickHouse, PostgreSQL-style data models, CLIs, event matching, and experiment tracking.

Rare Isle Fund — Principal Software Engineer

Apr 2019-Jun 2021 | Remote

- Built automated trading, research, analytics, and operations infrastructure from scratch for a hedge fund.
- Designed large-scale backtesting and strategy evaluation pipelines capable of processing millions of strategy variants daily.
- Built automated selection workflows, analytics dashboards, operational controls, and monitoring tools for deployed systems.
- Worked across Python, Pandas, Go, AWS, Grafana, Linux, DevOps, data pipelines, and production operations.

FocusVision — Senior Software Engineer

Nov 2015-Jun 2018 | Portland, OR

- Led backend integration work after acquisition, consolidating multiple Rails/PostgreSQL systems into a unified production platform.
- Managed and mentored junior engineers while delivering production SaaS features and integration work.
- Built and maintained SaaS features, database-backed workflows, admin tools, internal operations systems, and deployment automation.
- Worked across Rails, PostgreSQL, Ansible, Linux operations, data migration, production support, and release workflows.

ADDITIONAL EXPERIENCE

24tru — Software Engineer

Apr 2013-Jun 2015 | Lakewood, CO

- Built SaaS features for market research products, including video upload, video encoding, transcription workflows, and backend operations.
- Worked across Rails, PostgreSQL, DevOps, Linux, and production support in a small engineering team.

Tango Group — Software Developer

Feb 2012-Oct 2012 | Boulder, CO

- Implemented Ruby on Rails features, fixed bugs, wrote tests, and supported PostgreSQL-backed product development.

SELECTED SYSTEMS

Full-Stack Operations Workstation

Operator-facing real-time workstation for monitoring and controlling multiple accounts from one interface.

- Included live account state, risk limits, kill switches, flatten/cancel controls, fan-out logic, monitoring views, and operational explanations.

Data & Analytics Infrastructure

Ingestion, normalization, storage, query, and analysis workflows for structured event and market datasets.

- Built with PostgreSQL, ClickHouse, SQLite, Python, SQL, event-based data models, replayable logs, and query-heavy analytics workflows.

Research & Simulation Platform

Rust/Python infrastructure for replay, simulation, evaluation, backtesting, runtime workflows, evidence capture, and reproducible research.

- Supported strategy/runtime evaluation, historical replay, simulated workflows, artifact tracking, and operator review.

Automation & Internal Tools

CLIs, dashboards, browser extensions, background workflows, deployment tooling, monitoring dashboards, and local research systems.

- Focused on data-heavy operational workflows, alerts, event matching, insight extraction, and AI-assisted operator tooling.

TECHNICAL STACK

Languages: Rust, Python, Ruby, Go, TypeScript, SQL

Frontend: React, TypeScript, HTML, CSS, dashboards, browser extensions, admin tools

Backend: Ruby on Rails, FastAPI, APIs, WebSockets, background jobs, service architecture, CLI systems

Data: PostgreSQL, ClickHouse, SQLite, ETL, data modeling, event logs, analytics pipelines

Infrastructure: Kubernetes, Docker, AWS, Linux, Ansible, CI/CD, sysadmin, Grafana, monitoring

Domains: SaaS, internal tools, automation, analytics, research infrastructure, trading systems, data-heavy products